

SELF-EVOLUTION SHOULD NOT DEPEND ON HOW SKILLS ARE SPLIT: AN EXACT CERTIFICATE FOR SKILL-TAXONOMY REPRESENTATION INVARIANCE

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Paper under double-blind review

ABSTRACT

Self-evolving language-model agents increasingly use skill identities as control objects. In Skill Self-Play (Skill-SP), the sampled skill determines the next training-task proposal; in SkillRL, fresh dynamic IDs receive priority inside a fixed retrieval budget. This creates a representation-invariance question at the control surface itself: if two skill libraries expose the same semantic capabilities while differing only in how those capabilities are split, cloned, or grouped into package identities, should the induced self-evolution control change? Package identity acts before the downstream outcome is observed, so representation-dependent exposure changes which semantic regions receive control opportunity even when the semantic support relation is unchanged. We formulate *Skill-Taxonomy Representation Invariance* (STRI) as this system-level property and audit it in two released self-evolving skill systems. Skill-SP API-Bank Level-1 contains 183 multi-membership tasks among 314 released-validator-covered tasks, and exact-support package pruning still leaves 71 overlapping tasks. In SkillRL, exact-content skills with fresh dynamic IDs are accepted 12/12 times and alter the unique semantic retrieval set for 11/12 tested targets. For the exact mechanism analysis, we freeze Skill-SP’s independently defined context-by-package support matrix A and define $R^*(A) = \min_{w \geq 0, t \geq 0} \{t : 1 \leq Aw \leq t1\}$. We prove that $R^*(A) = 1$ exactly when package-only nonnegative mass can equalize additive semantic exposure, while $R^*(A) > 1$ is the tight minimum multiplicative distortion. Level-1 has $R^* = 2$, with the same value on frozen calibration and tool-disjoint heldout subsets. Two load-bearing negative regimes define the mechanism boundary: Level-3 has $R^* = 1$ with disjoint support, while released logical compilers yield 127/128 multi-membership tasks and still have $R^* = 1$ because umbrella support permits exact equalization. The mechanism is therefore non-equalizable support geometry rather than overlap prevalence. STRI-Cert packages first-party support provenance, the strongest same-information package-mass baseline, the exact certificate, and optional lower-bound witnesses into a fail-closed audit. Standard linear programming computes the certificate; the contribution is the representation-invariance question and its exact support-geometry boundary for the audited control class.

1 INTRODUCTION

Many self-evolving agents adapt through external state, and skills are increasingly used as the identities through which that state is sampled, retrieved, and updated. Recent systems make the identity itself a control variable: Skill-SP (Huang et al., 2026) samples a skill before generating a training task, while SkillRL (Xia et al., 2026) gives dynamically created skill IDs priority inside a fixed retrieval budget.

This architecture makes the *representation* of the skill library scientifically consequential. Consider two taxonomies that expose the same executable semantic capability. One stores a primitive once while another gives the same content a fresh package identity. Another split keeps generic and specific support regions separate. The primitive semantics can remain fixed while package-indexed

control changes which semantic regions receive future exposure. The invariance question therefore arises before downstream utility is measured: a self-evolution controller should have a well-defined semantic control surface under equivalent package representations.

We call that system-level property *Skill-Taxonomy Representation Invariance* (STRI). The exact theorem in this paper audits one concrete control class exposed by released Skill-SP: package-only, pre-context, nonnegative additive exposure over a frozen finite support matrix. This scope relation is central to the paper. STRI names the representation property; $R^*(A)$ is the exact certificate for the audited action class.

Prior work establishes that skill quality, redundancy, routing, and lifecycle governance matter. The unresolved question is whether an endogenous skill controller itself is invariant to semantics-preserving changes in package representation. We study that question asset-first on released control surfaces. The decisive reduction gives the baseline the complete frozen support matrix and arbitrary nonnegative package mass, so the scientific result cannot be attributed to a weak router, a dedup heuristic, or a particular learned optimizer.

Released Skill-SP. On API-Bank Level-1, 314 released tasks are covered by at least one released validator and 183 have multiple memberships. The release’s name+description Jaccard duplicate filter at threshold 0.33 misses all five observed specific–generic support-overlap pairs. Minimum-cardinality whole-package pruning can reduce the active library from 10 to 7 while preserving exact support, but 71 overlapping tasks remain. The residual is therefore not merely an exact-clone artifact.

Released SkillRL. In released template-mode retrieval, dynamic general skills are included before the remaining fixed $\text{top}_k=6$ budget is filled with static skills. Adding one exact-content fresh-ID copy at a time admits all 12 tested clones. Across 223 released ALFWorld task descriptions, 11/12 clone targets change the unique semantic retrieval set and 5/12 reduce the number of unique general-skill contents returned.

Skill-SP and SkillRL jointly establish the released-system representation-sensitivity claim N1. SkillRL is used for that independent identity-sensitivity phenotype. The exact support-geometry claims N2–N3 are audited on Skill-SP, where first-party support contracts expose the required matrix directly. A natural alternative mechanism is overlap prevalence. The high-overlap equalizable regime below is the decisive falsifier for that explanation.

Contributions.

1. **Representation-invariance problem.** We formulate STRI and establish released package-identity sensitivity in two independent self-evolving skill systems.
2. **Exact audited boundary.** For package-only pre-context additive exposure on a frozen support matrix, $R^*(A)$ is an exact iff certificate for equalizability. A global-singleton theorem gives an interpretable factor-2 lower bound when its support pattern is present.
3. **Mechanism falsifier.** Level-1 and its frozen splits have $R^* = 2$. Level-3 is equalizable, and the logical-compiler domain remains equalizable despite 127/128 multi-membership tasks. These negative regimes make support geometry, rather than overlap prevalence, the load-bearing explanation.
4. **Fail-closed audit protocol.** STRI-Cert binds independently defined support truth to the strongest same-information package-mass reduction, the exact certificate, and human-readable witnesses.

2 FROM PACKAGE IDENTITY TO SEMANTIC EXPOSURE

2.1 FROZEN SUPPORT MATRIX

Let $X = \{x_1, \dots, x_n\}$ be a finite set of semantic contexts or tasks with independently defined first-party support truth, and let $S = \{s_1, \dots, s_m\}$ be the current skill packages. Define the binary incidence matrix

$$A_{ij} = \mathbf{1}\{s_j \text{ supports } x_i\}.$$

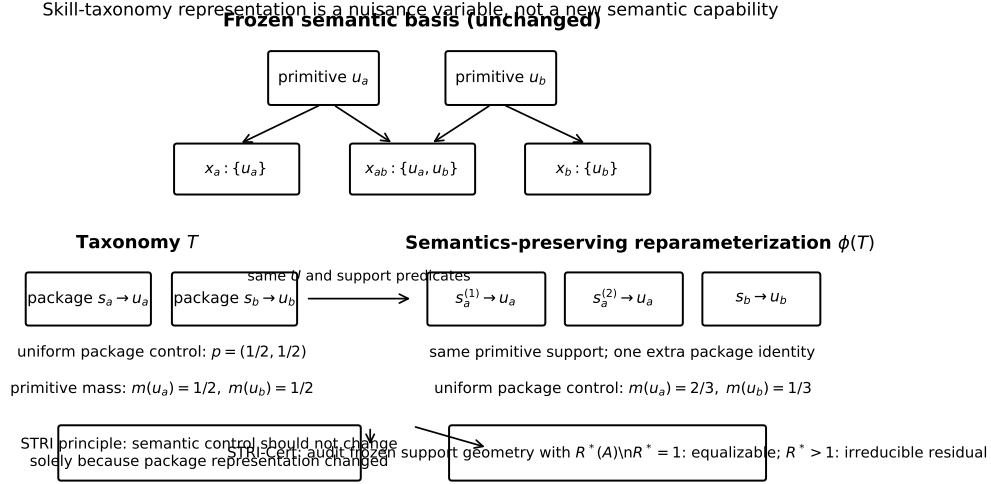


Figure 1: STRI separates semantic capability from package representation. The conceptual identity-split example leaves the primitive basis and support predicates unchanged, yet a package-indexed controller can change primitive mass solely because one primitive has two package identities. Exact clones motivate the nuisance channel; the decisive released residual in Sec. 4 survives clone reasoning and is characterized by non-equalizable partial-overlap support geometry.

The support definition is frozen before evaluating STRI. In the tool-call study, support comes from released Skill-SP validators. In the logical-reasoning study, tasks are generated from released compiler specifications, checked by fixed CSP/contract code, and cross-skill support is determined by the released compiler-alignment predicate. No language model or learned support estimator labels A in the N2–N3 audit.

We study package-only pre-context control. At a frozen state, the controller chooses nonnegative package mass $w \in \mathbb{R}_{\geq 0}^m$ before the semantic task to be generated exists. The additive eligibility exposure of context x_i is $e_i = (Aw)_i$. This is a control-opportunity quantity, not a task probability and not utility.

2.2 SAME-INFORMATION PRE-TASK CONTROLLERS REDUCE TO PACKAGE MASS

In released Skill-SP, skill selection occurs before task generation. At this decision point the action is one current package identity. For any fixed pre-task state z , every randomized same-information rule induces a categorical distribution over packages and therefore one nonnegative mass vector $w(z)$. The load-bearing baseline is arbitrary nonnegative package mass with access to the complete frozen support matrix. Text deduplication, exact-clone removal, pruning, uniform routing, and scalar retuning are weaker descriptive reductions.

The reduction is pointwise in the reviewed pre-task state z and respects the controller’s information timing: the baseline sees all frozen support information available for the audit while the future generated task remains unavailable at the pre-task action. This is the action class decided exactly by $R^*(A)$.

3 EXACT EQUALIZABILITY AND STRI-CERT

3.1 EXACT FINITE-SUPPORT CRITERION

The max/min exposure ratio is invariant to positive scaling of w . We normalize every covered context to exposure at least one and define

$$R^*(A) = \min_{w \geq 0, t \geq 0} t \quad \text{s.t.} \quad 1 \leq Aw \leq t1. \quad (1)$$

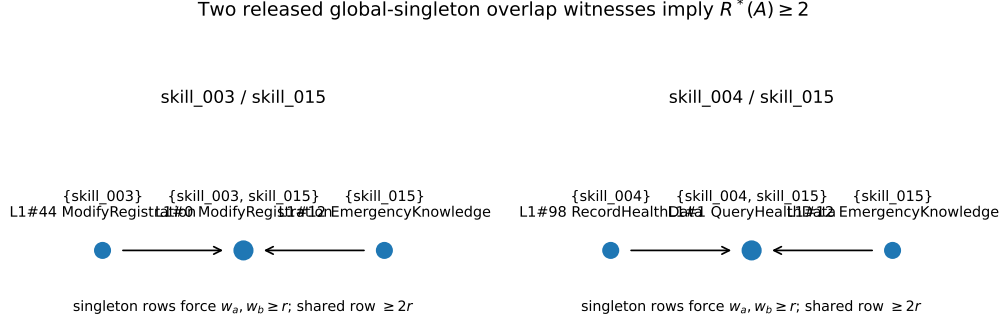


Figure 2: Two released global-singleton overlap witnesses. In each case, retaining positive exposure on the two singleton regions forces at least twice that minimum exposure on the shared region.

Theorem 1 (finite-support package-only equalizability). $R^*(A) = 1$ if and only if there exists a nonnegative package-only controller that gives exactly equal additive exposure to every covered context. If $R^*(A) > 1$, no such package weighting exists, and $R^*(A)$ is the tight minimum achievable max/min exposure ratio.

Proof. If exact equalization exists, rescale its positive common exposure to one, giving a feasible solution with $t = 1$. The constraints imply $t \geq 1$, so $R^*(A) = 1$. Conversely, $R^*(A) = 1$ implies $1 \leq Aw \leq 1$, hence $Aw = 1$. For any w with positive exposure on every covered row, let $r = \min_i (Aw)_i$ and rescale to $\tilde{w} = w/r$. Then $\min_i (A\tilde{w})_i = 1$ and the smallest feasible t is $\max_i (Aw)_i / \min_i (Aw)_i$. Minimizing over w yields exactly the tight multiplicative distortion. $\square$

The optimization itself is standard linear programming. The scientific content is the representation-invariance object that the LP exactly decides for a released self-evolution control surface.

3.2 HUMAN-READABLE FACTOR-2 WITNESS

Theorem 2 (global-singleton overlap lower bound). Suppose packages a, b each have a context supported by exactly that one package over the full frozen support universe, and another context is supported by both a and b , possibly with additional packages. If both singleton contexts retain positive exposure, then every nonnegative package mass has max/min exposure ratio at least two on the focal rows, and therefore $R^*(A) \geq 2$ for the full support matrix.

Proof. Let $r > 0$ be the minimum exposure among the two singleton contexts and the shared context. Since the first singleton support signature is exactly $\{a\}$, its exposure is $w_a \geq r$; likewise $w_b \geq r$. The shared context contains both a and b , and additional packages have nonnegative weight, so its exposure is at least $w_a + w_b \geq 2r$. Thus every w has focal max/min ratio at least two. For the same w , the full-matrix maximum is no smaller than the focal maximum and the full-matrix minimum is no larger than the focal minimum, so the full-matrix ratio is also at least two. Minimizing over w gives $R^*(A) \geq 2$. $\square$

The released Skill-SP Level-1 graph contains two independent instances, involving skill_003–skill_015 and skill_004–skill_015. The exact LP attains $R^* = 2$, so the lower bound is tight.

3.3 STRI-CERT

STRI-Cert is a fail-closed audit protocol. It uses standard linear programming as the exact decision subroutine:

1. freeze an independently defined context-by-package support matrix;
2. solve Eq. (1);

Table 1: Released-system evidence for package-identity representation sensitivity. The two systems expose different control channels; only Skill-SP supplies the support matrix used by STRI-Cert.

System	Control surface	Released evidence
Skill Self-Play (tool-call)	Pre-task skill sampling and skill-ID feedback/refinement	183 multi-membership Level-1 rows among 314 covered rows; the released text duplicate filter misses all 5/5 observed specific-generic support-overlap pairs; exact-support whole-package pruning still leaves 71 overlap rows.
SkillRL	Fixed-budget template-mode dynamic skill retrieval	12/12 exact-content fresh-ID clones are admitted; 11/12 clone targets change the unique semantic retrieval set over 223 released ALFWorld task descriptions; 5/12 reduce the number of unique general-skill contents.

Table 2: Reduction tournament on the Skill-SP tool-call support matrix. The strongest same-information reduction is arbitrary nonnegative global package weighting with access to the complete frozen support matrix.

Reduction	Outcome	Evidence
Released name+description Jac-card duplicate filter	Does not absorb	Misses all 5/5 observed specific-generic support-overlap pairs; similarity 0.1429–0.20 is below the released 0.33 threshold.
Exact semantic-clone deduplication	Does not absorb partial overlap	The positive regime uses distinct specific/generic packages with different unique support regions.
Minimum-cardinality whole-package pruning preserving exact support	Partial reduction only	Active packages drop from 10 to 7, but 71 overlap rows remain while exact support is preserved.
Arbitrary nonnegative global package weights	Exact residual	The exact optimum is $R^*(A) = 2$.
Global-singleton overlap witness	Closed-form lower bound	Two independent released witnesses imply $R^*(A) \geq 2$, and the exact LP attains 2.

3. if $R^* = 1$, report *package-only equalizable*;
4. if $R^* > 1$, report an *irreducible package-only representation residual* and its tight distortion;
5. search for globally valid singleton-overlap structures to provide interpretable lower bounds when available;
6. fail closed if support truth or optimization is invalid.

The strongest same-information reduction is built into the certificate: arbitrary nonnegative global package reweighting receives the complete frozen support matrix. Text deduplication, exact-clone removal, uniform routing, and fixed scalar retuning are weaker reductions.

4 RELEASED-SYSTEM EVIDENCE

4.1 SKILL-SP TOOL-CALL SUPPORT: POSITIVE REGIME

On API-Bank Level-1, 314 rows are accepted by at least one released Skill-SP validator and 183 have multiple memberships. STRI-Cert gives $R^* = 2.0$.

To test whether this is an artifact of fitting package weights to the complete table, we use a tool-disjoint split frozen before the later certificate analysis. Calibration uses AppointmentRegistration, ModifyRegistration, QueryHealthData, and RecordHealthData; heldout uses CancelRegistration, EmergencyKnowledge, QueryRegistration, and SymptomSearch. Calibration has 47 covered tasks, 33 multi-membership rows, and $R^* = 2$. Heldout has 52 covered tasks, 38 multi-membership rows, and again $R^* = 2$. Both independently contain the two globally valid factor-2 witness structures.

Table 3: Mechanism boundary across real first-party support regimes. High overlap is neither necessary nor sufficient for an irreducible package-only residual; the exact discriminator is $R^*(A)$.

Regime	Covered	Multi	Overlap	R^*	Certificate
API-Bank Level-1 full	314	183	58.3%	2.0	Residual
Level-1 calibration tools	47	33	70.2%	2.0	Residual
Level-1 tool-disjoint heldout	52	38	73.1%	2.0	Residual
API-Bank Level-3	34	0	0.0%	1.0	Equalizable
Logical compiler validation	128	127	99.2%	1.0	Equalizable

Table 4: Preregistered faithful Qwen3-8B dynamic pilot. Qualification is checked before any dynamic STRI witness may update scientific belief.

Source skill	Generated	Contract-valid	Required
skill.003	24	14	16
skill.004	24	5	16
skill.015	24	8	16

Formal outcome: `INCONCLUSIVE_PROPOSER_QUALIFICATION_FAILED`. The 72-generation bank supplies no dynamic STRI belief update and is not rerun.

The witness rows are directly inspectable. For `skill.003/skill.015`, L1#44 and L1#12 are singleton regions while L1#0 is shared. For `skill.004/skill.015`, L1#98 and L1#12 are singleton regions while L1#1 is shared. In each case Theorem 2 yields the same lower bound attained by the exact LP.

4.2 LEVEL-3: DISJOINT-SUPPORT NEGATIVE CONTROL

Level-3 is the first load-bearing mechanism control. Thirty-four rows are covered, every covered row has exactly one skill membership, and $R^* = 1$. Equal weights on the active packages equalize every covered task. The certificate therefore changes with support geometry rather than firing mechanically on a released taxonomy.

4.3 LOGICAL COMPILER DOMAIN: HIGH OVERLAP BUT NO RESIDUAL

The decisive mechanism control keeps overlap high while changing the support geometry. Skill-SP’s logical-reasoning release contains eight skill packages with programmatic compiler specifications. Fixed code generates a zebra puzzle from a compiler, verifies it with a CSP solver, and checks the sample against each package’s released compiler-alignment contract.

We freeze the author’s default compiler-validation sizes (3×3 , 4×4 , 5×5 , 6×6) and seeds 0, 1, 2, 3 for each of eight released skills, giving 128 source units. No language model is used. All 128/128 units compile and pass the author’s puzzle contract and uniqueness verifier; every task is then cross-evaluated against all eight released support specifications.

The support matrix is extremely overlapping: 127/128 tasks have more than one supported skill, with 38 distinct support patterns. Yet $R^* = 1$. The exact LP places weight one on `skill.008` and zero on the remaining packages because the released `skill.008` alignment contract supports every generated unit. This umbrella support makes the high-overlap matrix exactly equalizable.

Thus the mechanism is not “more overlap implies more representation dependence.” It is whether the support geometry admits an equalizing nonnegative package mixture.

4.4 SKILLRL: INDEPENDENT IDENTITY-SENSITIVITY WITNESS

SkillRL shows that the broader representation-sensitivity problem is not confined to one implementation. In released template-mode retrieval, fresh dynamic identities receive priority in a fixed general-skill budget. An exact-content clone under a fresh `dyn_NNN` ID is admitted for all 12 tested general skills. For 11/12 clone targets, the unique semantic retrieval set changes over 223 released

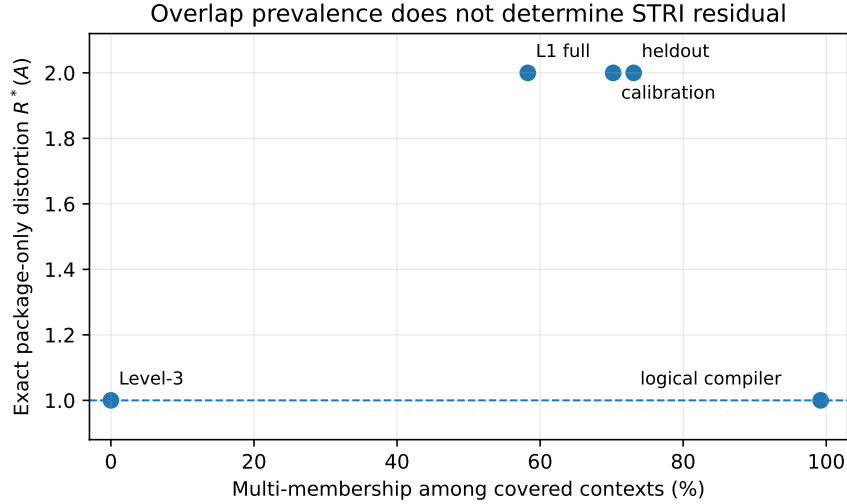


Figure 3: Overlap prevalence does not determine STRI residual. The three Level-1 regimes independently have $R^*(A) = 2$, while Level-3 is equalizable with disjoint support and the logical compiler domain is equalizable despite 127/128 multi-membership tasks.

task descriptions; for 5/12, the number of unique general contents decreases under the same `top_k=6` budget.

SkillRL supplies the independent released-system witness for N1. N2–N3 are evaluated on Skill-SP because its first-party support contracts expose the matrix needed by the exact certificate.

5 RELATION TO EXISTING WORK

Skill redundancy and clone detection. Prior work studies semantic skill clones, duplicate propagation, and clone-aware filtering (Zhu et al., 2026). STRI targets the remaining control question after exact-clone reasoning: the focal positive regime contains specific and generic packages with distinct unique support regions in addition to shared contexts.

Redundancy-aware lifecycle governance. Prior governance methods recommend, attribute, suppress, merge, or update redundant and environment-sensitive skills (Liu et al., 2026). STRI-Cert addresses the complementary property question: whether a frozen taxonomy admits representation-invariant package-only control under the same support information.

Skill safety and failures. Work on unsafe skill evolution and harmful agent skills demonstrates that learned or retrieved skill content can propagate functional or efficiency failures (Mao et al., 2026; Dong et al., 2026). STRI isolates a different variable by keeping the audited semantic support predicates fixed while testing package representation at the control surface.

Adaptive routing and evolving skills. ERSkill, Skill-SP, SkillRL, SHAPER, and related systems learn routers or co-evolve skill libraries (Chen et al., 2026; Huang et al., 2026; Xia et al., 2026; Wang et al., 2026). These works absorb any generic claim that adaptive routing or skill evolution is novel. They motivate STRI because package identities become endogenous actions or state, but they do not test invariance under a semantics-preserving reparameterization of that representation.

Compression and structured online learning. Skill-compression methods factor repeated procedural structure (Bai et al., 2026), while classical structured bandit/expert methods share information across similar fixed actions (Slivkins, 2011; Sen et al., 2018). STRI contributes the representation-invariance question for an endogenous artifact taxonomy and an exact certificate on independently defined semantic support.

6 DISCUSSION

STRI-Cert makes a representation property measurable at the same point where package identity enters control. When $R^*(A) > 1$, every package-only global weighting leaves residual exposure distortion on the audited support domain. When $R^*(A) = 1$, the support geometry admits an equalizing package mixture. The contrast between these regimes is the main scientific object.

The logical-compiler result is especially informative. Its 99.2% overlap rate is higher than the tool-call regime, yet umbrella support makes exact equalization possible. A redundancy count and a text-similarity score therefore predict the wrong boundary. The certificate instead follows the incidence geometry that the controller actually acts through.

This boundary suggests a concrete design direction. Controllers can factor decision state through semantic support cells or another representation that is invariant to package refactoring. The current paper supplies the diagnostic needed to tell when package-only control has an irreducible representation channel; future work can use that diagnostic to evaluate redesigned controllers.

7 LIMITATIONS

1. **Audited support domain.** STRI-Cert requires an independently defined finite support matrix. The exact result applies to the audited contexts represented in that matrix; learned or uncertain support estimation introduces a separate identification problem.
2. **Control quantity.** R^* certifies package-only additive exposure equalizability. It does not establish downstream task failure, regret, success degradation, or longitudinal utility; those outcomes correspond to the frozen unsupported claim U1.
3. **Optimization machinery.** Standard linear programming computes R^* . The paper’s contribution is the self-evolution representation question and its exact support-geometry boundary, not computational novelty over LP; computational novelty corresponds to frozen unsupported claim U2.
4. **Dynamic evidence.** The preregistered Qwen3-8B Skill-SP pilot completed all 72 frozen generations but failed the 16-valid-per-source proposer qualification gate with 14/24, 5/24, and 8/24 contract-valid outputs. A separately frozen SkillRL fixed-task bridge also lacks the preregistered equivalence margin and task-cluster power needed for a population-level dynamic conclusion. Neither result upgrades the paper to a general dynamic-effect claim; that boundary corresponds to frozen unsupported claim U3.
5. **Repair evaluation.** The certificate identifies an irreducible package-only representation channel and motivates representation-invariant controller designs. The present evidence evaluates the diagnostic boundary rather than a utility-improving repair policy.

8 CONCLUSION

Self-evolving agents should not change their semantic control process merely because the same capability is packaged differently. Released systems show that skill identity can nevertheless alter curriculum and retrieval control. The relevant mechanism is not redundancy in the abstract, but whether the support incidence geometry can be equalized by package-only control.

The quantity $R^*(A)$ turns this principle into an exact finite audit. Skill-SP tool-call support yields a tight factor-2 residual that replicates on tool-disjoint heldout tools. Two real negative regimes define the boundary: disjoint Level-3 support is equalizable, and a logical skill library remains equalizable despite nearly universal overlap. Together these results establish a representation-invariance problem, an exact support-geometry certificate, and a falsified alternative mechanism while keeping downstream utility claims explicitly separate.

AI USE STATEMENT

In this work, generative AI tools assisted with conceptual framing and hypothesis refinement; formulation and critique of mathematical claims and proof drafts; experiment and falsifier design; implementation and review of analysis, experiment-control, and manuscript-quality-assurance code; interpretation of intermediate results; primary-source literature triage; and preparation or editing of manuscript text, figures, tables, and references. Generative models were also used as the task proposer in the preregistered Qwen3-8B dynamic pilot summarized in the Limitations (Section 7); that pilot failed its frozen author-native qualification gate, and its generated bank remains outside the frozen N1–N3 claim set. Load-bearing numerical claims in the submitted narrow paper are bound to released first-party artifacts or deterministic/programmatic analysis outputs. The support certificate, theorem edge cases, transition logic, and paper-level numerical/citation constraints were exercised by executable tests; related-work boundaries and bibliographic metadata were checked against primary or official sources. AI-assisted outputs were not treated as independent scientific authority. We take responsibility for the final content of this work, including text, claims, code, and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

The main paper states the finite-support assumptions and complete proofs needed for the two certificate theorems, identifies the released-system support regimes used for every load-bearing result, and reports the exact positive and negative mechanism boundaries. The supplementary material will include the STRI-Cert implementation, first-party logical-compiler support protocol, frozen result/contract receipts, source ledger, figure-generation scripts, and machine-readable claim-to-evidence checks. All reported dynamic qualification thresholds were frozen before outcomes and the failed Qwen3-8B bank is retained as a transparent non-result rather than rerun or retuned.

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